

Git Collaboration & Troubleshooting Guide

This document outlines the standard workflow for repository management, team collaboration, and resolving common Git errors.

1. Repository Initialization

For creators setting up a new project.

- [cite_start]**Create README:** Create a basic documentation file: `echo "# test22" >> README.md` [cite: 2]
 - [cite_start]**Initialize Git:** Start a new Git repository: `git init` [cite: 3]
 - [cite_start]**Stage & Commit:** Add the README and create the first save point: `git add README.md && git commit -m "first commit"` [cite: 4, 5]
 - [cite_start]**Set Branch:** Ensure the default branch is named 'main': `git branch -M main` [cite: 6]
 - [cite_start]**Connect to Remote:** Link your local folder to GitHub: `git remote add origin https://github.com/` [cite: 7]
 - [cite_start]**Initial Push:** Upload your code: `git push -u origin main` [cite: 8]
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2. Team Collaboration Workflow

Steps for contributors joining an existing project.

Step 1: Access & Setup

- [cite_start]**Grant Access:** The repository creator must add team members as contributors in the repository settings [cite: 9].
- [cite_start]**Clone the Repo:** Once added, download the project: `git clone <repo-url>` [cite: 10, 11].

Step 2: Feature Branching

- [cite_start]**Create Branch:** Create a personal branch to work safely: `git branch Arsalan` [cite: 12, 13].
- [cite_start]**Switch Branch:** Move into your new branch: `git checkout Arsalan` [cite: 14].
- [cite_start]**Save Changes:** Stage your files and commit them: `git add the_file.py` && `git commit -m "Arsalan's modules"` [cite: 15, 16].
- [cite_start]**Push Branch:** Upload your branch to GitHub: `git push -u origin Arsalan` [cite: 17].

3. Merging Changes to Main

[cite_start]How to combine your work with the main project branch. [cite: 18, 19]

1. [cite_start]**Switch to Main:** `git checkout main` [cite: 20]
2. [cite_start]**Pull Latest:** Get recent updates from others: `git pull origin main` [cite: 21]
3. [cite_start]**Merge:** Combine your feature branch: `git merge Arsalan` [cite: 22]
4. [cite_start]**Final Push:** Update the main branch on GitHub: `git push origin main` [cite: 23]

4. Common Errors & Resolutions

Error Type	Description	Fix / Command
Merge Conflict	[cite_start]Same lines modified in two branches[cite: 25, 26].	[cite_start]Edit markers (<<<<<<< , =====), then <code>git add</code> and <code>git commit</code> [cite: 30-40].
Non-fast-forward	[cite_start]Remote <code>main</code> has updates you don't have[cite: 41, 42].	[cite_start] <code>git pull origin main --rebase</code> followed by <code>git push</code> [cite: 46, 47].
Uncommitted Changes	[cite_start]Local work prevents branch switching [cite: 48-52].	<code>git add .</code> [cite_start]&& <code>git commit</code> OR use <code>git stash</code> [cite: 54, 55].
Unrelated Histories	[cite_start]Merging two different repositories[cite: 56, 57].	[cite_start] <code>git merge <branch> --allow-unrelated-histories</code> [cite: 60].
Merge Aborted	[cite_start]Conflict prevents merge from starting[cite: 61, 63].	[cite_start] <code>git merge --abort</code> and clean your directory [cite: 65-67].
Permission Denied	[cite_start]Error 403: No write access [cite: 68-71].	[cite_start]Verify collaborator status or check login credentials[cite: 73, 75].

Error Type	Description	Fix / Command
Deleted Branch	[cite_start]Trying to merge a branch deleted on GitHub [cite: 76-78].	[cite_start]Merge locally or re-push the branch using <code>git push</code> [cite: 80, 81].
Detached HEAD	[cite_start]You are on a commit, not a branch [cite: 82-84].	[cite_start]Switch back to a branch: <code>git switch main</code> [cite: 85, 86].

Pro Tips

- [cite_start]**Commit Often**: Always commit or stash your work before switching branches [cite: 87, 88].
- [cite_start]**Stay Updated**: Always pull the latest changes before you start merging [cite: 89].
- [cite_start]**Clean History**: If merging is too messy, try `git rebase` for a linear history [cite: 90].